

Statistics II: Introduction to Inference

Module 2: Estimation

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Suppose we want to infer about some properties of a population (which is modeled as F_{θ}), based on a random sample from the population $X_1, \dots, X_n$. So, we assume $X_1, \dots, X_n \stackrel{i.i.d.}{\sim} F_{\theta}$. Any property of the population F_{θ} , can be characterized by a function of θ . In particular, assume that the property of interest be $\psi(\theta)$.

Recall that, any function of the random samples $X_1, \dots, X_n$, say $T_n = T(X_1, \dots, X_n) = T(\mathbf{X})$ is called a statistic. For a particular realization of $X_1, \dots, X_n$, say, $x_1, \dots, x_n$, the value of the statistic $T(\mathbf{x}) = t_n$ is a realization of the statistic T_n .

Definition 1 (Estimator). Suppose a statistic $T(\mathbf{X})$ is used to estimate $\psi = \psi(\theta)$, then $T(\mathbf{X})$ is called an **estimator** of $\psi(\theta)$. For a particular realization of $X_1, \dots, X_n$, say, $x_1, \dots, x_n$, the value of the estimator $T(\mathbf{x}) = t_n$ is called an **estimate** of $\psi(\theta)$.

Note: Like the random sample, a statistic (in particular, an estimator) $T(\mathbf{X})$ is also a random variable. The distribution of $T(\mathbf{X})$ is called the *sampling distribution* of $T(\mathbf{X})$. As the distribution of $\mathbf{X}$ depends on θ , it is natural to expect that the distribution of $T(\mathbf{X})$ also depends on θ .

Example 1: Let $X_1, \dots, X_n$ be a random sample of size n from $\text{Binomial}(m, \theta)$ distribution, with $m = 20$. Consider the following three estimators of θ :

$$T_1(\mathbf{X}) = \frac{1}{m} \bar{X}_n, \quad T_2(\mathbf{X}) = \sqrt{\frac{S_n^2}{m} + \frac{\bar{X}_n^2}{m^2}}, \quad \text{and} \quad T_3(\mathbf{X}) = \frac{1}{m} X_1.$$

We can obtain the sampling distribution of these statistics analytically, as well as using simulation.

- For different choices of θ , the sampling distributions of $T_1(\mathbf{X})$, $T_2(\mathbf{X})$ and $T_3(\mathbf{X})$ (calculated based on $n = 15$ random samples) are described in the following histograms (see Figure 1).
- In fact, it is possible to find the distribution of (squared) distance of T_j from θ , i.e., $D_{j,\theta}\{T(\mathbf{X}) - \theta\}^2$. For different choices of θ , the distributions of $D_{j,\theta}(\mathbf{X})$ (calculated based on $n = 15$ random samples) are described in Figure 2.
- From Figure 1, observe that, based on the choice of θ , the sampling distributions of T_1 , T_2 and T_3 change. The figures clearly indicate the following:
 1. T_1 and T_3 neither overestimate nor underestimate θ on an average.
 2. T_2 tends to overestimate θ . The amount of bias decreases as θ increases.
 3. From Figure 2, it is clear that the distribution of T_1 is much more concentrated around θ compared to T_2 and T_3 .
- Therefore, based on these behavior of the sampling distributions T_1 seems to be the best estimator of θ among these three proposed estimators.

From the above example, it is evident that the distribution of $D_{\theta}(\mathbf{X}) = \{T(\mathbf{X}) - \theta\}^2$ provides a clear idea of how close is the distribution of $T(\mathbf{X})$ compared to θ . To summarize the distribution of $D_{\theta}(\mathbf{X})$ in a scalar, one may consider its expectation (or, mean), which is the mean squared error.

Definition 2 (Mean-Squared Error (MSE)). Let $T(\mathbf{X})$ be an estimator of $\psi(\theta)$. The MSE of $T(\mathbf{X})$ in estimating $\psi(\theta)$ is given by

$$\mathcal{MSE}_{\psi(\theta)}(T_n) = \mathcal{MSE}_{\psi(\theta)}(T(\mathbf{X})) = E_{\theta} \left[\{T(\mathbf{X}) - \psi(\theta)\}^2 \right].$$

An estimator of $\psi(\theta)$ is the **best** if it minimizes the MSE in estimating $\psi(\theta)$ (for all θ).

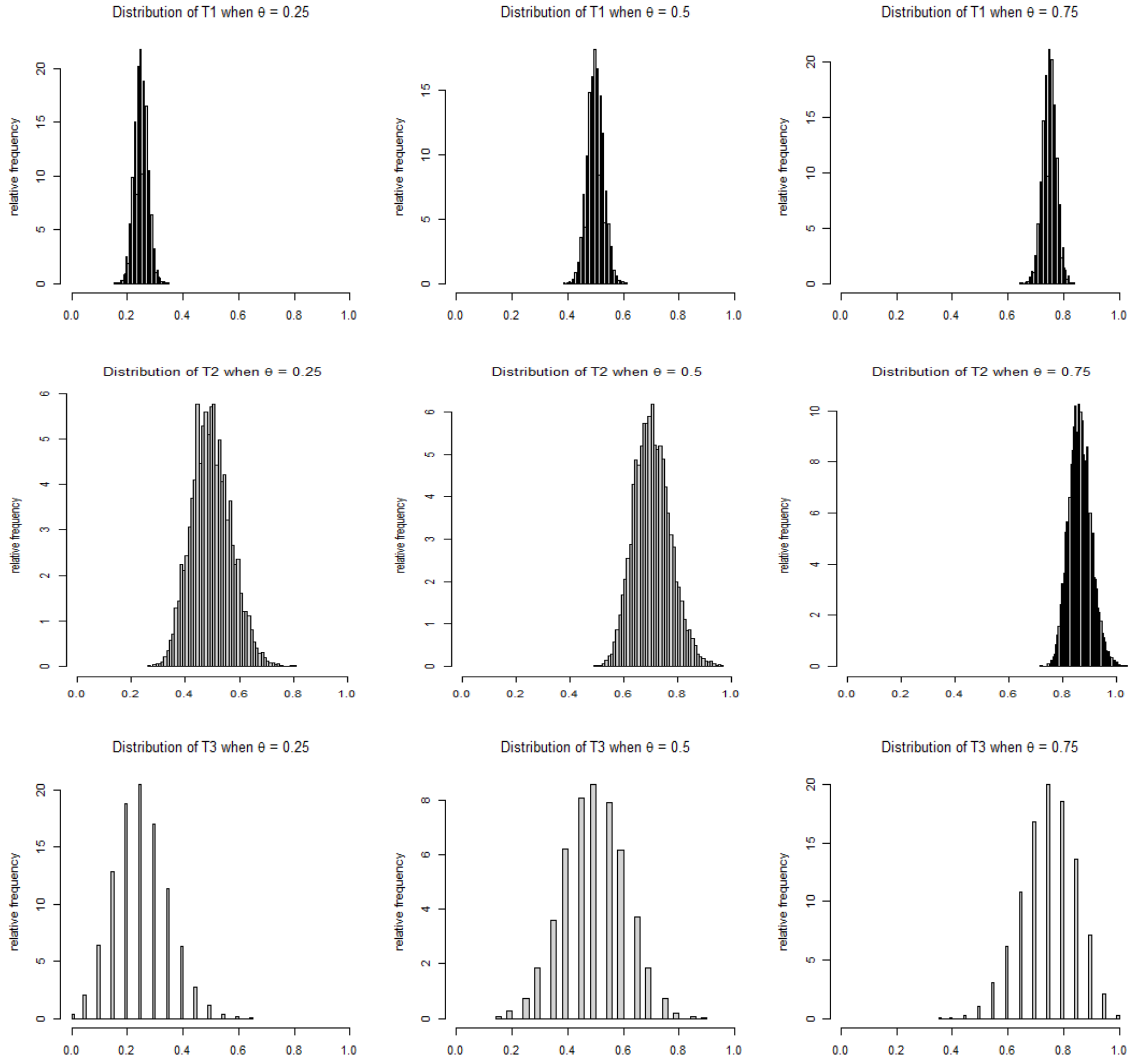


Figure 1: Sampling distributions of $T_1(\mathbf{X})$ (top panel), $T_2(\mathbf{X})$ (middle panel) and $T_3(\mathbf{X})$ (bottom panel) for 3 choices of θ , viz. $\theta = 0.25$ (left panel), $\theta = 0.5$ (middle panel) and $\theta = 0.75$ (right panel).

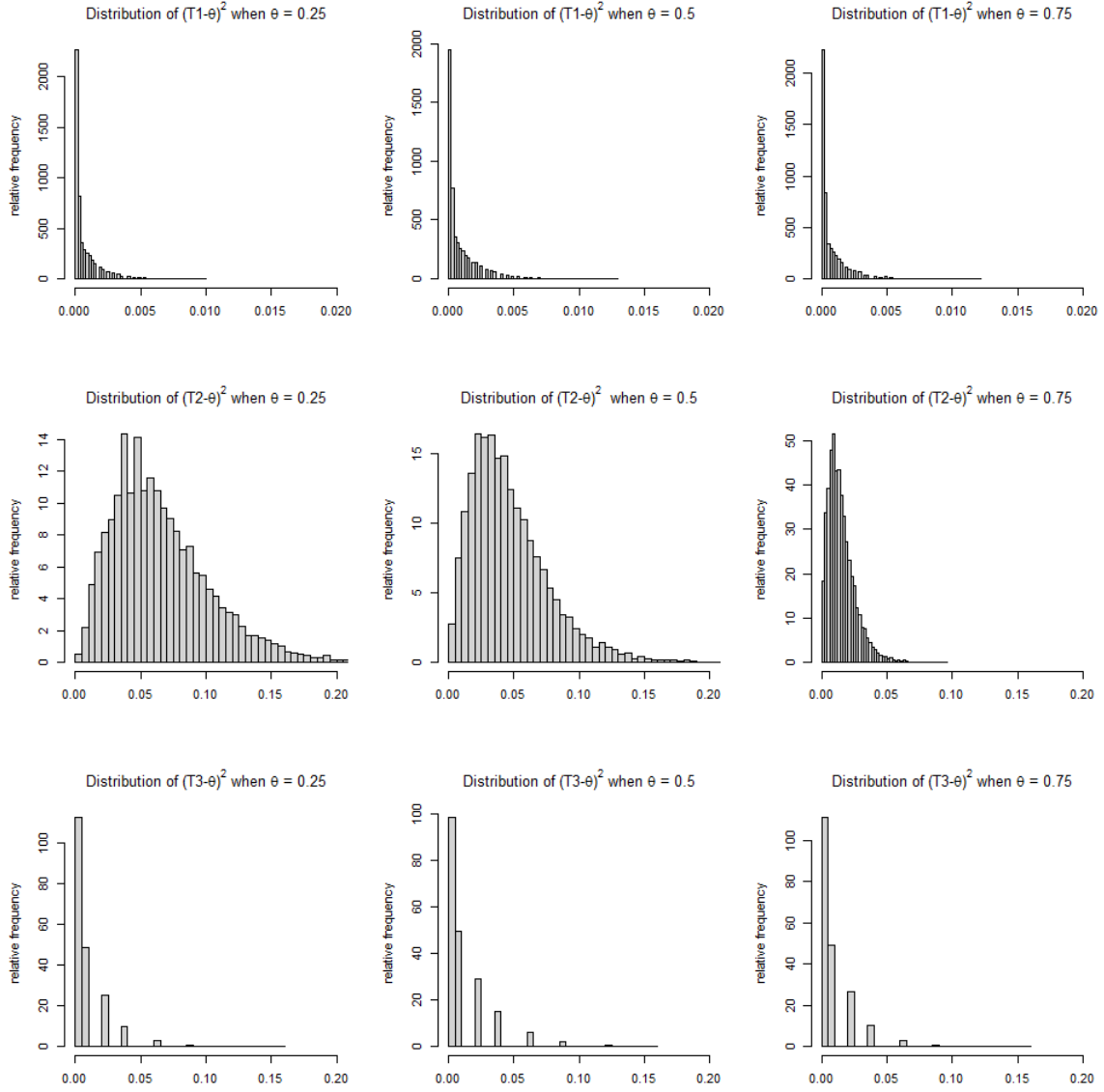


Figure 2: Distributions of $D_{1,\theta}(\mathbf{X})$ (top panel), $D_{2,\theta}(\mathbf{X})$ (middle panel) and $D_{3,\theta}(\mathbf{X})$ (bottom panel) for 3 choices of θ , viz. $\theta = 0.25$ (left panel), $\theta = 0.5$ (middle panel) and $\theta = 0.75$ (right panel).

Remark 1. Unfortunately, it is not usually possible to find the estimator with lowest possible MSE among the class of all estimators of $\psi(\boldsymbol{\theta})$. Therefore, we restrict our search to a subclass of the estimators and find the best estimator (estimator with lowest MSE) within that subclass. The subclass we consider is the class of *unbiased estimators* of $\psi(\boldsymbol{\theta})$, say $\mathcal{U}_{\psi(\boldsymbol{\theta})}$.

Definition 3 (Unbiased Estimator). *Let $T(\mathbf{X})$ be an estimator of $\psi(\boldsymbol{\theta})$. Then the bias in estimating $\psi(\boldsymbol{\theta})$ by $T(\mathbf{X})$ is defined as*

$$\mathcal{B}_{\psi(\boldsymbol{\theta})}(T_n) = E_{\boldsymbol{\theta}}(T(\mathbf{X})) - \psi(\boldsymbol{\theta}).$$

*An estimator $T(\mathbf{X})$ is called **unbiased** if $\mathcal{B}_{\psi(\boldsymbol{\theta})}(T_n) = 0$ for all $\boldsymbol{\theta}$.*

Remarks 2.

1. If $\mathcal{B}_{\psi(\boldsymbol{\theta})}(T_n) > 0$ (or, < 0) then $T(\mathbf{X})$ tends to overestimate (or, underestimate) $\psi(\boldsymbol{\theta})$ on an average.
2. It is easy to see that the MSE of T_n in estimating $\psi(\boldsymbol{\theta})$ is the sum of two components, the squared bias of T_n in estimating $\psi(\boldsymbol{\theta})$, and the variance of T_n , i.e.,

$$\mathcal{MSE}_{\psi(\boldsymbol{\theta})}(T_n) = \mathcal{B}_{\psi(\boldsymbol{\theta})}^2(T_n) + \text{var}_{\boldsymbol{\theta}}(T_n)$$

Therefore, in a pool of unbiased estimators, the estimator with smallest variance has the smallest MSE.

Definition 4 (Relative Efficiency). *Let $T_1(\mathbf{X})$ and $T_2(\mathbf{X})$ be two unbiased estimators of $\psi(\boldsymbol{\theta})$. Then the relative efficiency of T_1 and T_2 is*

$$\mathcal{RE}_{T_1, T_2}(\psi(\boldsymbol{\theta})) = \frac{\text{var}_{\boldsymbol{\theta}}(T_1)}{\text{var}_{\boldsymbol{\theta}}(T_2)}.$$

If $\mathcal{RE}_{T_1, T_2}(\psi(\boldsymbol{\theta})) < 1$ (or, > 1) then the estimator T_1 is preferred (or, not preferred) over T_2 .

Definition 5 (UMVUE). *An estimator of $\psi(\boldsymbol{\theta})$, $T(\mathbf{X})$, is called a **uniformly minimum variance unbiased estimator** (UMVUE) for $\psi(\boldsymbol{\theta})$ if*

- (i) $E_{\boldsymbol{\theta}}\{T(\mathbf{X})\} = \psi(\boldsymbol{\theta})$ for all $\boldsymbol{\theta}$, and
- (ii) for any estimator $T'(\mathbf{X})$ with $E_{\boldsymbol{\theta}}\{T'(\mathbf{X})\} = \psi(\boldsymbol{\theta})$, $\text{var}_{\boldsymbol{\theta}}\{T(\mathbf{X})\} \leq \text{var}_{\boldsymbol{\theta}}\{T'(\mathbf{X})\}$ for all $\boldsymbol{\theta}$.

Remarks 3.

1. UMVUE, if exists, is unique.
2. For a certain problem of estimation, the UMVUE can have higher MSE compared to a biased estimator. Thus, the UMVUE is not necessarily the estimator with lowest MSE (see Example 4 below).

How to find the UMVUE? One way to find the UMVUE is through the *complete-sufficient statistics* (CSS).

Definition 6 (Sufficient Statistic). *Let $\mathbf{X} = \{X_1, \dots, X_n\}$ be a random sample from the distribution $F_{\boldsymbol{\theta}}$. A statistic $T(\mathbf{X})$ is a **sufficient statistic** for $\boldsymbol{\theta}$, if the conditional distribution of the random sample $\mathbf{X}$ given $T(\mathbf{X})$ does not depend on $\boldsymbol{\theta}$.*

Definition 7 (Complete Family of Distributions). *Let $T(\mathbf{X})$ be a statistic. Then the distribution of T is called **complete** if $E_{\boldsymbol{\theta}}(g(T)) = 0$ for all $\boldsymbol{\theta}$ implies $P_{\boldsymbol{\theta}}(g(T) = 0) = 1$ for all $\boldsymbol{\theta}$. A statistic $T(\mathbf{X})$ is called **complete**, if its distribution is complete in the above sense.*

Result 1. *Let $T(\mathbf{X})$ be complete-sufficient statistic (CSS), and there exists a function g such that $E_{\boldsymbol{\theta}}[g(T(\mathbf{X}))] = \psi(\boldsymbol{\theta})$ for all $\boldsymbol{\theta}$, i.e., $g(T(\mathbf{X}))$ is an unbiased estimator of $\psi(\boldsymbol{\theta})$. Then $g(T(\mathbf{X}))$ is the UMVUE of $\psi(\boldsymbol{\theta})$.*

Example 2: Let $X_1, \dots, X_n$ be a random sample from $\text{uniform}(0, \theta)$. Suppose it is known that the highest order statistics $X_{(n)} = \max\{X_1, \dots, X_n\}$ is complete-sufficient.

1. Verify that $E_\theta(X_{(n)}) = n\theta/(n+1)$, for all θ , which implies

$$E_\theta \left[\frac{n+1}{n} X_{(n)} \right] = \theta.$$

2. Therefore, by the above result $T(\mathbf{X}) = (n+1)X_{(n)}/n$ is the UMVUE of θ .

How to identify a complete-sufficient statistic? Usually it is not easy to obtain a complete-sufficient statistic (beyond the scope of this course). However, when the class of distributions under consideration belongs to the exponential family, then there is a simple way out.

Definition 8 (Exponential family). *A family of pmfs of pdfs is called a d -parameter exponential family if it can be expressed as*

$$f_\theta(\mathbf{x}) = \exp \left\{ H(\mathbf{x}) + A(\theta) + \sum_{i=1}^k C_i(\theta) T_i(\mathbf{x}) \right\}, \quad \theta \in \Theta \subseteq \mathbb{R}^d.$$

Here $H, T_1, \dots, T_k$ are real valued functions of $\mathbf{x}$, not depending on θ . Further, $A(\theta), C_1(\theta), \dots, C_k(\theta)$ are real-valued functions of θ , not depending on $\mathbf{x}$.

Result 2. Let $X_1, \dots, X_n$ be a random sample from a distribution with pmf or pdf f_θ , $\theta \in \Theta \subseteq \mathbb{R}^d$, and let Θ contain an open subset of $\mathbb{R}^d$ which belongs to an exponential family with $d \leq k$. Then the statistic $\mathbf{T}(\mathbf{X})$ is jointly complete-sufficient for θ , where

$$\mathbf{T}(\mathbf{X}) = (T_1(\mathbf{X}), \dots, T_k(\mathbf{X})).$$

Note: A one-one function of a complete-sufficient statistic is also a complete-sufficient statistic.

Example 3: Let $X_1, \dots, X_n$ be a random sample from $\text{normal}(\mu, \sigma^2)$ distribution.

1. Express the joint pdf of $X_1, \dots, X_n$ in the form of exponential family.
2. Hence show that $(\sum_{i=1}^n X_i, \sum_{i=1}^n X_i^2)$ is jointly complete-sufficient.
3. Using the above result find the UMVUEs of μ and σ^2 .

Example 4: Let $X_1, \dots, X_n$ be a random sample from $\text{Exp}(\lambda)$ distribution with mean $1/\lambda$.

1. Verify that the sample mean $\bar{X}_n$ is an unbiased estimator of $1/\lambda$.
2. Also verify that the variance of $\text{var}_\lambda(\bar{X}_n) = 1/(n\lambda^2)$.
3. Show that $\bar{X}_n$ is UMVUE of $1/\lambda$.
4. Next consider another estimator $T(\mathbf{X}) = (n+1)^{-1} \sum_{i=1}^n X_i$.
5. Show that $T(\mathbf{X})$ has a negative bias of magnitude $1/(n+1)$. Further, $\text{var}_\lambda(T(\mathbf{X})) = n/\{(n+1)^2\lambda^2\}$.
6. Hence, show that MSEs of $\bar{X}_n$ and $T(\mathbf{X})$ in estimating $1/\lambda$ are $1/(n\lambda^2)$ and $1/\{(n+1)\lambda^2\}$, respectively. Thus, $T(\mathbf{X})$ is a biased but better estimator than the UMVUE.